LIU HE

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RESEARCH INTERESTS

Agentic Techniques for Computer Architecture and Systems

- Architecture-aware agent reasoning and optimization knowledge transfer for kernel performance portability across heterogeneous architectures.

Computer Architecture Design with Advanced and Emerging Technologies

- 3D-integrated heterogeneous memory architecture design for large language model acceleration.

EDUCATION

Tsinghua University, Beijing, China **09/2023 – Present**
Ph.D. in Electronic Science and Technology
Advisor: Associate Professor Hongyang Jia

Tsinghua University, Beijing, China **08/2019 – 06/2023**
B.Eng. in Electronic Engineering

WORK EXPERIENCE

Biren Technology, Shanghai, China **06/2025 – 07/2025**
Intern

- Agent design for CUDA kernel generation and performance optimization.

Intel Labs China, Beijing, China **06/2022 – 12/2022**
Research Intern

- Branch predictor design for general-purpose processors.

PUBLICATIONS

Conferences:

1. **L. He**, F. Zhou, C. Peng, S. Dong, Z. Zhang, H. Yang, Y. Liu, G. Zhang, H. Jia, "SHyLA: 3D-Stacked NVM-DRAM Hybrid LLM-Inference Architecture Exploiting Data and Memory Heterogeneity," *2026 IEEE/ACM 53rd Annual International Symposium on Computer Architecture (ISCA)*, Raleigh, NC, USA, June 2026.
2. **L. He**, Y. Wang, Z. Huang, S. Fan, C. Tang, S. Zhang, L. Lei, H. Yang, Y. Liu, H. Jia, "Exploring Approximation and Dataflow Co-Optimization for Scalable Transformer Inference Architecture on the Edge," *2024 IEEE 37th International System-on-Chip Conference (SOCC)*, Dresden, Germany, September 2024.
3. Y. Wang, S. Dong, Z. Huang, Y. You, **L. He**, H. Yang, Y. Liu, H. Jia, "HyC-LoRA: Memory Efficient LoRA Fine-tuning with Hybrid Activation Compression," *2025 8th Conference on Machine Learning and Systems (MLSys)*, Santa Clara, CA, USA, May 2025.

Journal:

1. **L. He**, Z. Huang, Y. Wang, S. Fan, C. Tang, S. Zhang, L. Lei, H. Yang, Y. Liu, H. Jia, "SASDenSebLE: A Compact Vision Transformer Inference Architecture With Saturation-Approximate Softmax Dataflow Enabling Sequence-Parallelism Boosted Layer-Fusion Execution," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2025.

TEACHING EXPERIENCE

Teaching Assistant, Tsinghua University **Autumn Term 2025 – 2026**
Course: Analysis and Design of Digital Very Large-Scale Integrated (VLSI) Systems
Teacher: Prof. Hongyang Jia

Teaching Assistant, Tsinghua University **Summer Term 2023 – 2024**
Course: Integrated Circuit Design Whole-Process Practice
Teacher: Prof. Yongpan Liu

Teaching Assistant, Tsinghua University **Spring Term 2023 – 2024**
Course: FPGA Based Design Methodology For Advanced Digital System
Teacher: Prof. Hongyang Jia

Teaching Assistant, Tsinghua University **Spring Term 2022 – 2023**
Course: Modern Computer Architecture
Teacher: Prof. Yongpan Liu & Prof. Hongyang Jia

AWARDS

Comprehensive Excellence Scholarship for Graduate Students - Tsinghua University	2026
Comprehensive Excellence Scholarship for Graduate Students - Tsinghua University	2025
Comprehensive Excellence Scholarship for Undergraduate Students - Tsinghua University	2022
Comprehensive Excellence Scholarship for Undergraduate Students - Tsinghua University	2021
Academic Excellence Scholarship for Undergraduate Students - Tsinghua University	2020

SHORT BIO

Liu He is a fourth-year Ph.D. student in the Department of Electronic Engineering at Tsinghua University, advised by Associate Professor Hongyang Jia. She received her B.Eng. degree in Electronic Engineering from Tsinghua University in 2023. Her research spans computer architectures for AI and AI-driven methods for systems and architecture design. Her earlier work focused on efficient AI inference through 3D-integrated heterogeneous memory architectures and approximation–dataflow co-optimization. Building on this background, she now investigates how AI agents can reason about architectural constraints and apply expert optimization knowledge to system design. Her current work focuses on extracting optimization principles from expert-designed kernels and adapting them across heterogeneous architectures to enable kernel performance portability.